

Sri Lanka Institute of Information Technology



Software Engineering | SE2030

Year 02 Semester 01 – 2025

Group ID – 55 (2025-Y2-S1-MLB-B3G2-05)

FINAL REPORT

LEARN - MATE

Web-based Educational Institute Management System

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Submission date – 25/10/2025

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1.0 Introduction

1.1 Project Overview

"Learn Mate" is a fully functional, Java-based web application developed to streamline the administrative and academic operations of a modern educational institute. The system successfully transitions the institute from a reliance on error-prone, manual processes (like paper files and spreadsheets) to a centralized, automated, and efficient digital platform. It serves as a single source of truth, connecting management, staff, teachers, students, and parents.

1.2 Objectives

The project set out to achieve the following objectives, all of which were successfully met:

- **To Help Management:** Provided the Director with a comprehensive dashboard displaying real-time analytics on student attendance, academic performance, and fee collection status.
- **To Reduce Admin Work:** Automated critical tasks such as student enrollment, timetable scheduling, and fee record management, significantly reducing manual effort and errors.
- **To Support Teachers:** Delivered a dedicated platform for teachers to mark attendance, enter and manage student grades, upload learning materials, and communicate with students and parents.
- **To Inform Students and Parents:** Created an intuitive portal for students and parents to access timetables, view academic results, pay fees online, and receive important announcements.
- **To Support Students Better:** Implemented an alert system that proactively notifies Student Support Officers of students with low attendance or declining grades, enabling timely intervention.

1.3 Target Users or Stakeholders

The system was designed for and successfully caters to six primary user roles:

1. **Director:** For high-level oversight and decision-making.
2. **Admin Staff:** For managing all operational data and user accounts.
3. **Teachers:** For academic delivery and student assessment.
4. **Students:** As the core beneficiaries for accessing their academic life.
5. **Parents:** For monitoring their child's progress and handling payments.
6. **Student Support Officer:** For student welfare and proactive support.

1.4 Scope and Limitations

- **In Scope:** The development included all six core modules: User Management, Class & Timetable Management, Attendance Tracking, Exams & Marks Management, Fee Management, and Communication & Notifications.
- **Out of Scope & Limitations:** As planned, the system is a single-institute solution. It does not include a native mobile app (though it is responsive), advanced e-learning features like live classes, or complex financial accounting beyond fee tracking. The notification system's reliability is dependent on external email/SMS APIs.

2.0 Requirements

2.1 Functional Requirements

- **FR1: User and Access Management:** Implemented with a secure login system and role-based access control (RBAC). Users can only access authorized features.
- **FR2: Timetable Management:** Admins can create and publish schedules. Students and teachers view personalized timetables.
- **FR3: Attendance Tracking:** Teachers can mark attendance per class. Records are viewable by students, parents, and authorized staff.
- **FR4: Exam and Marks Management:** Teachers can enter marks. Students and parents can view results securely through their portals.
- **FR5: Fee Processing:** Admins manage fee records. Parents can view balances and make online payments via an integrated gateway.
- **FR6 & FR7: Communication and Automated Notifications:** The system allows for targeted announcements and sends automated notifications for events like new results, fee reminders, and announcements.
- **FR8: Report Generation:** The Director's dashboard includes graphical and tabular reports on attendance, performance, and finances.

2.2 Non-functional Requirements

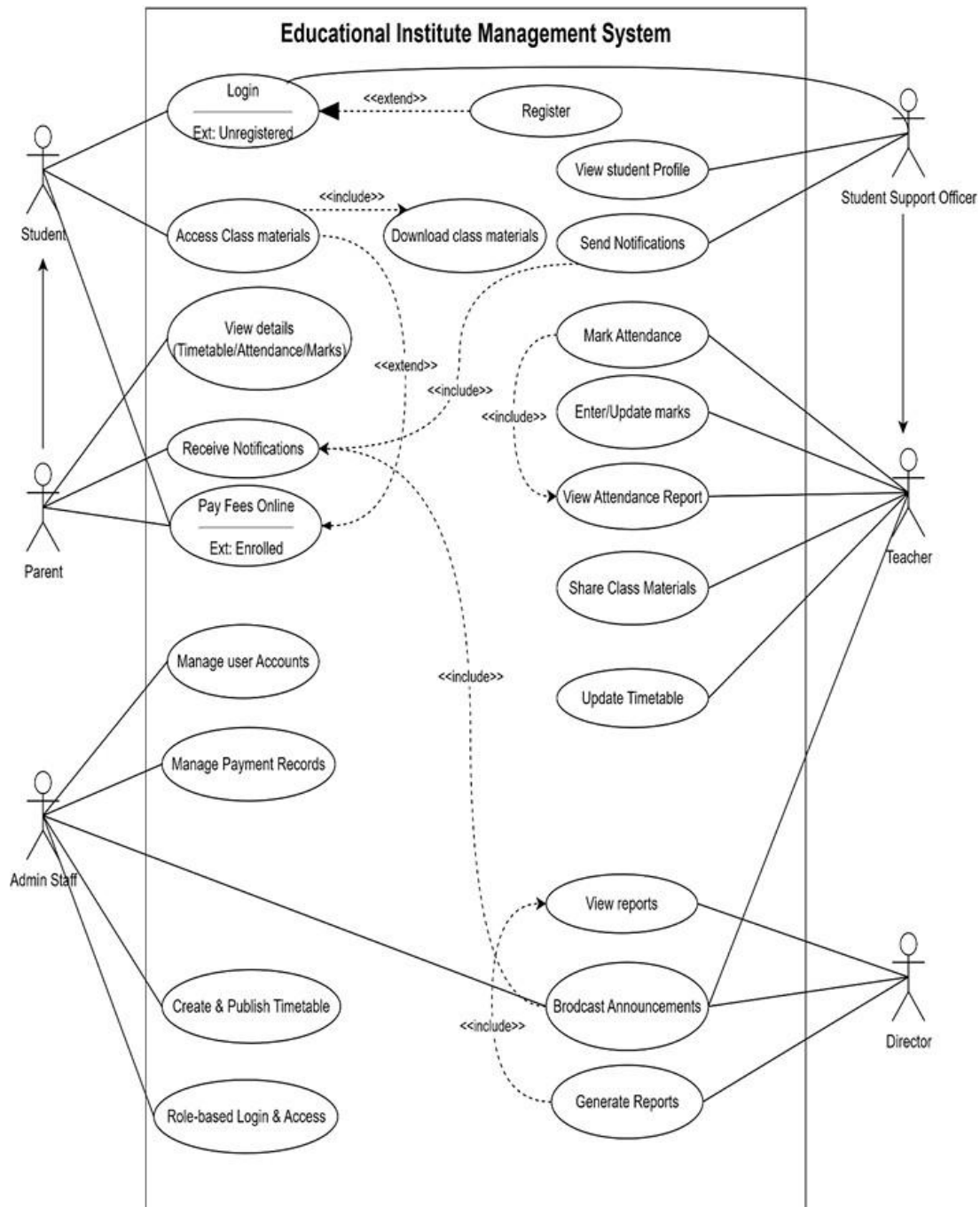
- **Performance:** The application is responsive, with key pages loading well under the 3-second target due to efficient backend design and database indexing.
- **Usability:** The interface was designed with a consistent, clean, and intuitive layout, requiring minimal training for new users. It underwent several usability refinements based on feedback.
- **Security:** Implemented Spring Security for authentication. Passwords are encrypted using BCrypt. Authorization checks are performed on all endpoints to ensure data isolation between users.
- **Compatibility:** The web application is fully functional on modern browsers (Chrome, Firefox, Edge) and is responsive across desktop, tablet, and mobile screens.

2.3 Constraints or Assumptions

- **Technology Constraint:** The system was built as required using Java (Spring Boot) and MySQL.
- **Assumption:** Users are expected to have a stable internet connection as the system is entirely web-based with no offline mode.
- **Constraint:** The project was developed within the strict 14-week academic timeline, which guided our Agile approach and feature prioritization.

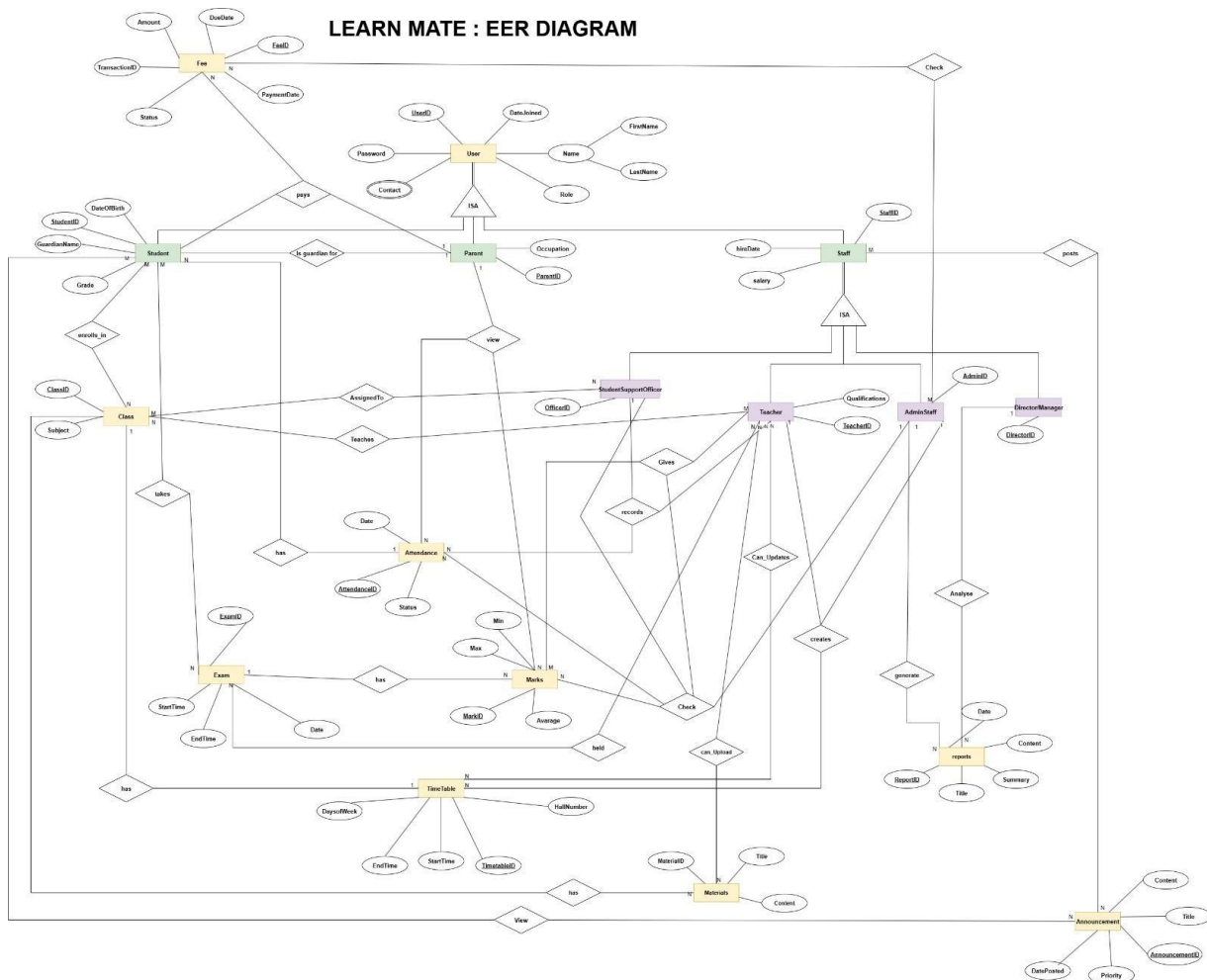
3.0 Design

3.1 Use Case Diagram



Description: The final Use Case Diagram illustrates all interactions between the six actors (Director, Admin, Teacher, Student, Parent, Support Officer) and the system's core use cases. It clearly shows how, for example, only a 'Teacher' can 'Mark Attendance' and 'Enter Marks', while a 'Student' can 'View Timetable' and 'View Marks'.

3.2 Database Design / ER Diagram



Description: The database was designed with a normalized schema to ensure data integrity and eliminate redundancy. Key entities include User (with a role field for polymorphism), Student, Teacher, Class, Subject, Attendance, Exam, Mark, FeeTransaction, and Notification. The ER diagram shows the one-to-many and many-to-many relationships, such as between Student and Class (Many-to-Many via an enrollment table.)

3.3 UI Sketches/Screenshots and Descriptions

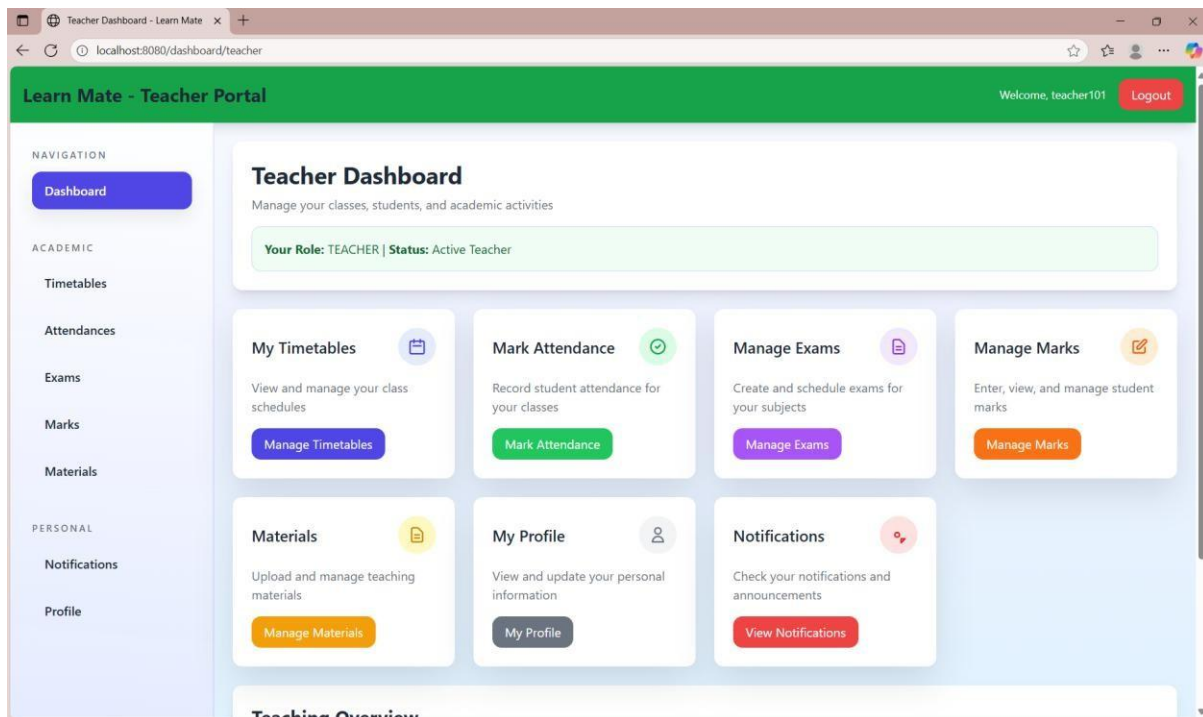


Figure 3.3.4: Teacher Dashboard

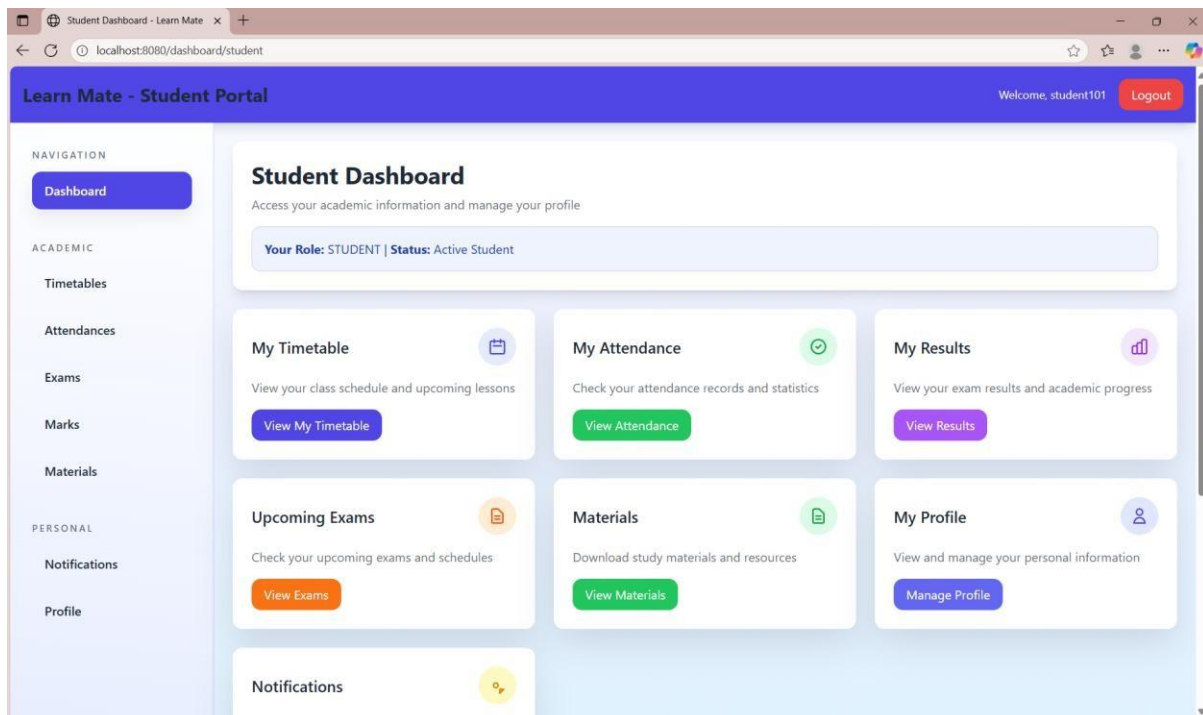


Figure 3.3.5: Student Portal

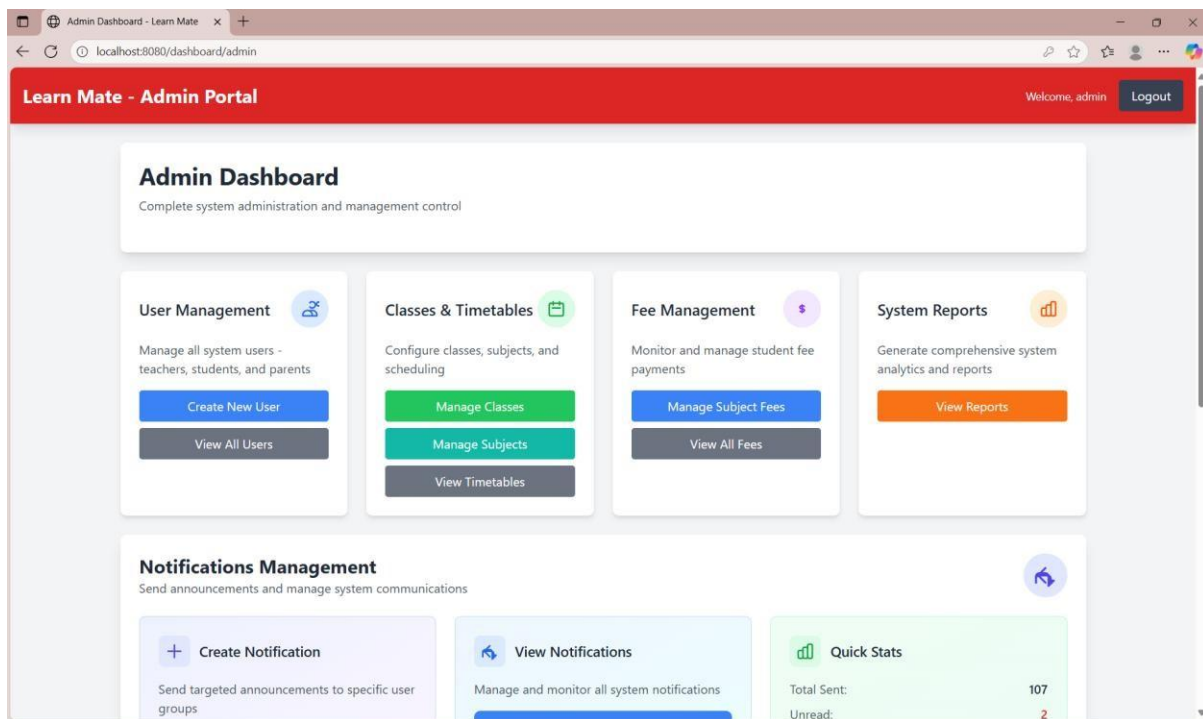


Figure 3.6: Director's Analytics Dashboard

- **Figure 3.4: Teacher Dashboard:** A screenshot showing the teacher's landing page with quick links to mark attendance, enter marks, and upload materials. The clean layout and clear navigation are evident.
- **Figure 3.5: Student Portal:** A screenshot of a student's view, displaying their weekly timetable, recent announcements, and a list of their published marks.
- **Figure 3.6: Director's Analytics Dashboard:** A screenshot showing summary widgets and charts for overall attendance percentage, class-wise performance, and fee collection status.

3.4 System Architecture

Our 'Learn Mate' system is built on a classic **Layered Architecture**, which is common for Spring Boot applications and follows the **Model-View-Controller (MVC)** design pattern.

- **Presentation Layer (View):** This is the frontend, built with Thymeleaf and HTML/CSS, which is responsible for displaying the user interface.
- **Business Logic Layer (Controller/Service):** Our `@Controller` classes handle incoming web requests, and our `@Service` classes contain the core business logic, coordinating actions and enforcing rules.
- **Data Access Layer (Model/Repository):** This layer consists of our `@Entity` classes (the Model) and our `JpaRepository` interfaces, which handle all communication with the MySQL database."

4.0 Implementation

4.1 Tools and Technologies Used

Backend: Java 17, Spring Boot (MVC, Data JPA, Security), Maven

Frontend: Thymeleaf, HTML5, CSS3, JavaScript, Bootstrap

Database: MySQL, Hibernate ORM

Development Tools: IntelliJ IDEA, Git/GitHub for version control, MySQL Workbench

Other: BCrypt for password encryption, SMTP for email notifications

4.2 Key Features Developed

All six major functions from the proposal were successfully implemented as fully working modules:

1. A robust **User Management** system with RBAC.
2. A dynamic **Class & Timetable Management** module.
3. A streamlined **Attendance Tracking** system.
4. A comprehensive **Exams & Marks Management** portal.
5. An integrated **Fee Management** system with online payment simulation.
6. A real-time **Communication & Notifications** hub.

4.3 Screenshots of Core Functions

The screenshot shows the 'Create Timetable' form in the Learn Mate application. The form is titled 'Create Timetable' and is located in the 'ACADEMIC' section of the sidebar. The form includes a 'Schedule Details' section with 'Day:-' and 'Time:-' fields. Below this, there are three required fields: 'Subject *', 'Teacher *', and 'Room *'. The 'Subject *' and 'Teacher *' fields are dropdown menus with the placeholder text 'Select a subject' and 'Select a teacher' respectively. The 'Room *' field is a text input with the placeholder text 'Enter room number/name'. At the bottom right of the form, there are two buttons: 'Cancel' and 'Add'.

Learn Mate - Create Timetable

localhost:8080/timetables/create

Learn Mate

Dashboard Logout

NAVIGATION

Dashboard

ADMINISTRATION

Users

Subjects

Classes

Reports

ACADEMIC

Timetables

Attendances

Exams

Marks

Materials

Create Timetable

Schedule Details

Day:-

Time:-

Subject *

Select a subject

Teacher *

Select a teacher

Room *

Enter room number/name

Cancel Add

Screenshot 4.3.1: timetable managing option for teacher

The screenshot shows the 'Mark Attendance' form in the Learn Mate application. The form is titled 'Mark Attendance - Grade 10' and is located in the 'ACADEMIC' section of the sidebar. The form includes a 'Select Class' dropdown menu with 'Grade 10' selected. Below this, there is a date and time display: 'Saturday, October 25, 2025 • 4 students'. To the right of this display, there are two buttons: 'Mark All Present' and 'Mark All Absent'. Below the date and time display, there is a 'Notes (Optional)' section with a text input field. At the bottom of the form, there is a table with the following columns: '#', 'Student Name', 'Student ID', and 'Attendance'. The table contains three rows of student data.

Learn Mate - Mark Attendance

localhost:8080/attendances/mark/5?date=2025-10-25

Learn Mate

Dashboard Logout

NAVIGATION

Dashboard

ACADEMIC

Timetables

Attendances

Exams

Marks

Materials

PERSONAL

Notifications

Profile

Select Class > Grade 10

Mark Attendance - Grade 10

Saturday, October 25, 2025 • 4 students

Teacher: teacher101

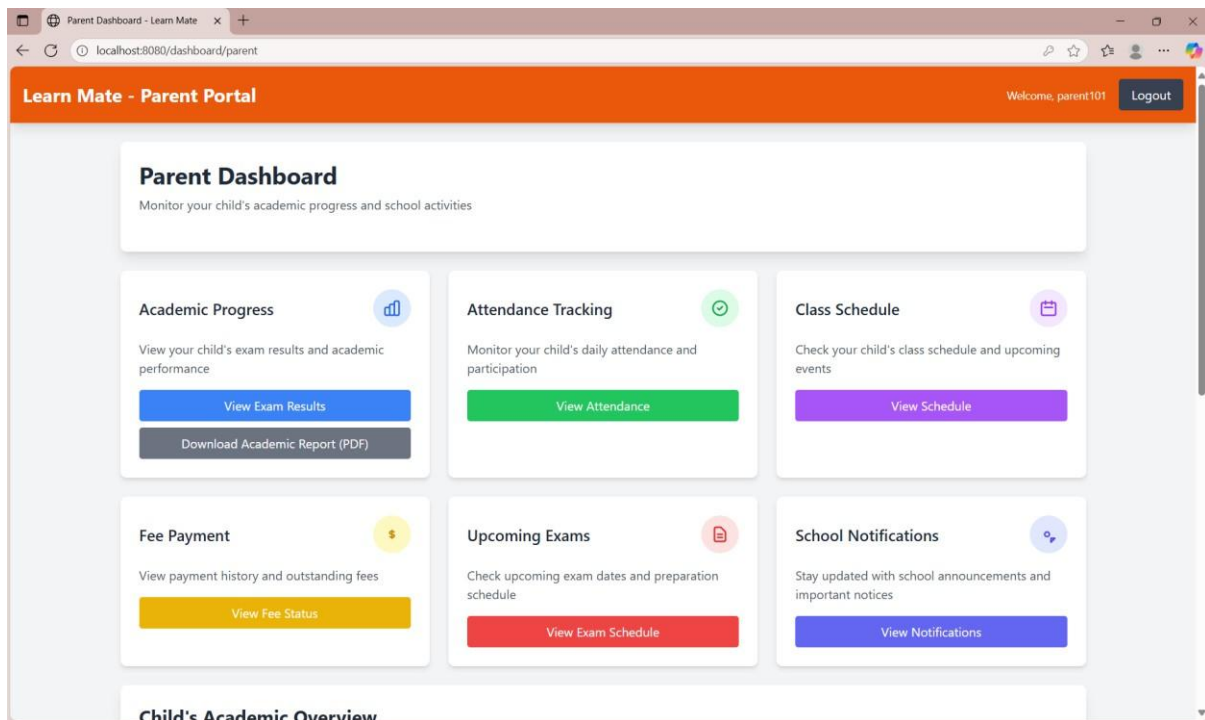
Mark All Present Mark All Absent

Notes (Optional)

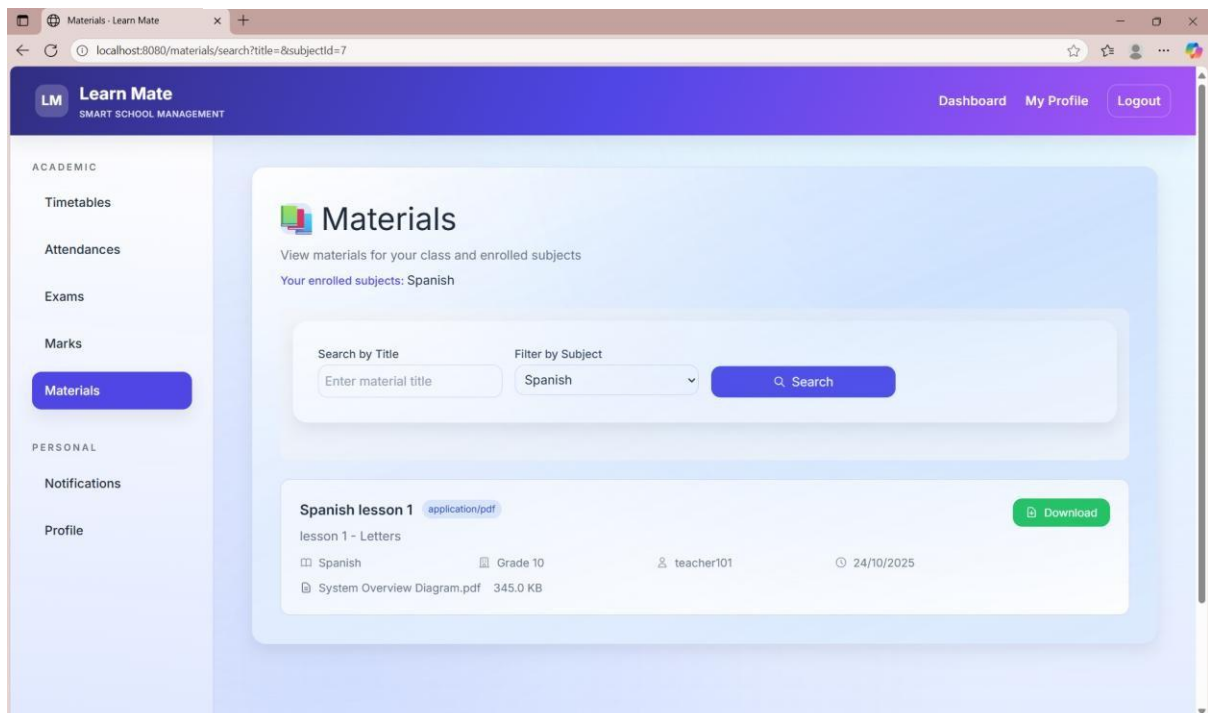
Add any notes about this attendance session...

#	Student Name	Student ID	Attendance
1	 student1 student1@gmail.com	17	☐ Absent
2	 Mike Johnson mike.johnson@gmail.com	22	☐ Absent
3	 student18 student18@gmail.com	35	☐ Absent

Screenshot 4.3.2: Attendance marking option



Screenshot 4.3.3: Dashboard view for the parent



Screenshot 4.3.4: Materials managing page

5.0 Project Management

5.1 Agile Approach and Sprint Summary

We adopted an Agile Scrum methodology, organizing our work into four 2-3 week sprints.

Sprint 1 (Weeks 4-5): Focused on foundational setup, database design, and developing the User Management and core authentication system.

Sprint 2 (Weeks 6-8): Developed the Timetable, Attendance, and Marks modules. Conducted internal testing and created the basic UI structure.

Sprint 3 (Weeks 9-10): Focused on Fee Management, Notifications, and integrating all modules. Prepared for and delivered the Progress Presentation.

Sprint 4 (Weeks 11-13): Dedicated to bug fixing, UI/UX refinement, implementing the Strategy Pattern for search, writing the final report, and preparation for the final viva.

5.2 Task Distribution Among Team Members

The six major functions were owned by individual team members as per the initial proposal:

Dulakshika A.L.S.H: User Management & Security

Bandara R. M. G. L: Class & Timetable Management

Dilhani W. P. K. A: Attendance Tracking

Dahanayake L.K: Exams & Marks Management

Devinda S. U. V: Fee Management

Disanayaka D.M.C.N: Communication & Notifications

All members contributed to frontend development, testing, and documentation for their respective modules.

5.3 Development Phases or Timeline

The project adhered to the proposed timeline through the four sprints mentioned above, culminating in the Final Submission in Week 14.

6.0 Conclusion & Future Work

6.1 Summary of Achievements

The "Learn Mate" project was a resounding success. We delivered a fully functional, robust, and user-friendly web application that meets all initial objectives and requirements. The system effectively centralizes institute operations, automates key tasks, and provides valuable insights to all stakeholders, thereby solving the core problems identified at the project's inception.

6.2 Challenges Faced

Database Relationship Complexity: Designing an efficient and scalable database schema to handle complex relationships (e.g., students in multiple classes with multiple subjects) was initially challenging.

Integration Hurdles: Ensuring seamless data flow between independently developed modules (e.g., linking a student from the User module to a class in the Timetable module

UI Consistency: Maintaining a consistent look and feel across different pages developed by multiple team members was addressed by using a shared Bootstrap template and regular UI reviews.

6.3 Suggestions for Improvement or Extension

Live Chat Feature: Implement a real-time messaging system between teachers, students, and parents.

Mobile Application: Develop a dedicated mobile app for faster, push-notification- driven access.

Advanced Analytics: Incorporate AI/ML models to predict student performance and provide deeper insights to the management.

E-learning Integration: Add features like video lecture hosting, online assignments, and quiz modules

7.0 Individual Contribution, Teamwork & Lessons Learned

Member Name & ID	Role & Contribution	Challenges & Solutions	Key Lessons Learned
Dulakshika A.L.S.H (IT24100300)	Role: Lead Backend Developer, User Management. Contribution: Designed and implemented the core database schema, Spring Security configuration, and the entire User Management module.	Challenge: Implementing secure role-based access control across the entire application. Solution: Thoroughly studied Spring Security, created custom security expressions, and conducted rigorous endpoint testing.	Learned the critical importance of security from the ground up and how to effectively manage complex object-relational mappings with Hibernate.
Bandara R. M. G. L (IT24100237)	Role: Timetable & Class Module Developer. Contribution: Developed the algorithms and interfaces for creating,	Challenge: Handling scheduling conflicts and ensuring data consistency when updating a published timetable.	Gained deep practical experience in backend algorithm
	updating, and displaying dynamic timetables for classes, teachers, and students.	Solution: Implemented server-side validation checks before any update and used database transactions.	design and the importance of data integrity checks.
Dilhani W. P. K. A (IT24100263)	Role: Attendance Module Developer. Contribution: Built the attendance marking system, including the UI for teachers and the backend logic for recording and retrieving attendance records.	Challenge: Making the attendance marking process fast and intuitive for teachers with large classes. Solution: Designed a single-page, AJAX-enhanced interface that allows for quick marking without page reloads.	Improved skills in frontend- backend integration (AJAX) and user-centric design thinking

Dahanayake L.K (IT24100314)	Role: Exams & Marks Module Developer. Contribution: Implemented the functionality for creating exams, entering marks, and calculating/generating student result reports.	Challenge: Ensuring the accuracy of calculated fields (e.g., averages, totals) and preventing duplicate mark entries. Solution: Implemented business logic validation in the service layer and unique constraints in the database.	Learned to never trust user input and to always validate business rules on the server side.
Devinda S. U. V (IT24100283)	Role: Fee Management Module Developer. Contribution: Developed the fee record management for admins and the payment portal for parents, including payment status tracking.	Challenge: Simulating a secure and reliable payment flow without integrating a real payment gateway for academic purposes. Solution: Created a mock payment service that mimics the steps of a real transaction while logging all actions.	Understood the flow and key considerations of e-commerce transactions and the importance of transaction logs.
Disanayaka D.M.C.N (IT24100239)	Role: Notification & Communication Lead, Design Pattern Specialist. Contribution: Built the announcement and notification system. Championed and implemented the Strategy Pattern for the materials search feature.	Challenge: Designing a notification system that was not intrusive yet effective. Refactoring search logic to use a design pattern. Solution: Used a non-blocking approach for sending emails. Successfully refactored the code, making it more maintainable as per the OCP.	Learned the practical benefits of design patterns in making code extensible and clean. Appreciated the value of team communication in a modular project.

8.0 Overall Team Reflection

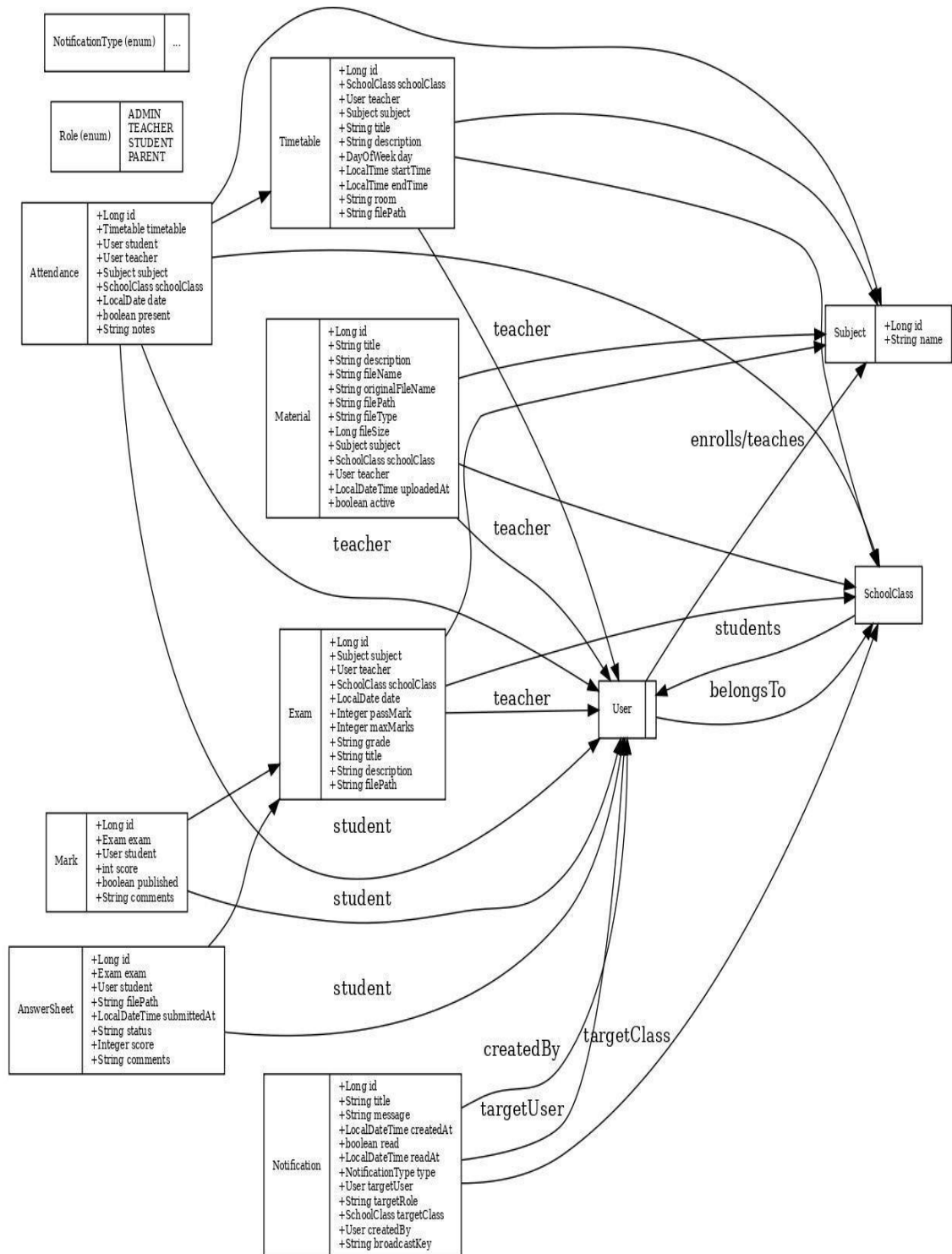
The project went very well in terms of adhering to the Agile methodology, meeting deadlines, and maintaining good team communication through weekly sync-ups. Our use of Git/GitHub for version control was effective. If we were to improve, we would start integration testing even earlier in the process to catch module-interfacing issues sooner.

9.0 References

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10.0 Appendix

- **GitHub Repository Link:** <https://github.com/loch214/learn-mate-system.git>
- **Final Class Diagram:**



- **Sprint Logs:**

Sprint Summary Plan 1

Task ID	User Story (Summary)	Priority	Assigned Member	Status
US01	Manage user accounts (CRUD)	High	IT24100300	Complete
US02	Create and publish timetables	High	IT24100237	Complete
US03	Mark student attendance	High	IT24100263	Complete
US04	Enter and manage marks	High	IT24100314	Complete
US05	View personalized timetable	High	IT24100237	Complete
US06	Access grades and marks	High	IT24100314	Complete
US07	View child's academic results	Medium	IT24100314	Complete
US08	Track child's attendance	Medium	IT24100263	Complete
US09	Manage fee records	High	IT24100263	Complete
US10	View fee balances and pay online	High	IT24100283	Complete
US14	View own attendance history	Medium	IT24100263	Complete
US15	View daily schedule	Medium	IT24100237	Complete

Sprint Summary Plan 2

Task ID	User Story (Summary)	Priority	Assigned Member	Status
US11	View real-time dashboard	Medium	IT24100300	Complete
US12	Broadcast announcements	High	IT24100239	Complete
US13	Receive notifications	High	IT24100239	Complete
US16	Send messages to parents	High	IT24100239	Complete
US19	Generate summary reports	Medium	IT24100300	Complete
US20	Receive school announcements	High	IT24100239	Complete
US17	Get alerts for low attendance	Medium	IT24100263	Complete
US18	Get alerts for failing grades	Medium	IT24100314	Complete
US21	Access student's complete profile	Medium	IT24100300	Complete
US22	View historical academic data	Medium	IT24100314	Complete
US23	Send private messages	Medium	IT24100239	Complete
US24	View teacher activities overview	Medium	IT24100300	Complete